

VYVAR Install (EN)

VYVAR release bundle - English install guide (RELEASE-2)

Preview bundles ship a **bundled Python 3.12 runtime** (R1). Your system Python is not used. User data (config, database, Archive, catalogs) lives in a separate **data directory** from the install folder.

1. Download and unpack

1. Download `VYVAR--win64.zip` or `VYVAR--linux-x64.tar.gz` from GitHub Releases on the public `VYVAR-release` repository.
2. Verify SHA256 against `SHA256SUMS` in the same release.
3. Unpack to a permanent location, e.g.:
 - Windows: `C:\Program Files\VYVAR\`
 - Linux: `~/apps/vyvar/`
4. Paths with spaces are supported.

2. First launch

Windows: double-click `VYVAR.bat` or run from cmd:

```
...  
cd C:\Program Files\VYVAR\VYVAR-preview-20260723-win64  
VYVAR.bat  
...
```

Linux:

```
...  
chmod +x vyvar.sh  
./vyvar.sh  
...
```

On first launch the data directory is created automatically:

Platform Default data directory

Windows `%LOCALAPPDATA%\VYVAR`

Linux `~/local/share/vyvar`

Override before launch: set environment variable `VYVAR\_DATA\_DIR` to any writable path.

The initializer creates `Archive/`, `CalibrationLibrary/`, catalog folders, an empty `vyvar.sqlite3`, `config.json` from template, and `NEXT\_STEPS.txt`.

3. Health check (selftest)

```
...
```

```
VYVAR.bat --selftest
./vyvar.sh --selftest
````
```

Prints Python version, platform, data directory, key dependency versions, and imports every compiled science module. Exit code 0 means the install is healthy.

## 4. Build catalogs (required - never shipped)

Catalogs are **always** built by the user (R2). Use the scripts from a dev checkout (or copy outputs into your data directory):

Catalog Script Typical output under data dir

```

Gaia DR3 SQLite `GAIA_DR3/build_gaia_catalog.py` `GAIA_DR3/vyvar_gaia_dr3.db`
Blind indexes `GAIA_DR3/build_blind_index.py` `GAIA_DR3/gaia_triangles_*.pkl`
VSX local `VSX/vsx_make.py` `VSX/vyvar_vsx_local_v2.db`
Exoplanets `exoplanets/exoplanet_make.py` `exoplanets/vyvar_exoplanet_local.db`
```

Point paths in Settings or edit `config.json` in the **data directory** after building.

## 5. Equipment setup (DB Explorer)

Open **Database Explorer** in the UI. Add:

1. **Location** (lat/lon/alt)
2. **Telescope**
3. **Equipment** (camera, filters, gain)

For **OSC cameras**, set **BAYERMASK** on the equipment record (required for correct debayer/channel handling).

## 6. Upgrade

1. Close VYVAR.
2. Replace the **install directory** with the new bundle (or unpack over it).
3. Do **not** delete the data directory - config, DB, and catalogs are preserved.

## 7. Uninstall

1. Delete the install directory.
2. Optionally delete the data directory if you no longer need observations/catalogs.

## 8. Troubleshooting

Issue Action

```

App will not start Run `--selftest`; check antivirus / Windows SmartScreen (unsigned binaries)
No stars / no comps Catalog paths empty or wrong - build catalogs (section 4)
Linux import errors Requires glibc >= 2.39 (Ubuntu 24.04 baseline)
Wrong data location Set `VYVAR_DATA_DIR` before launching
```

## License

See `LICENSE` in the install directory (proprietary; same terms as the dev repository).